

Notes: Tetlock (RFS, 2011)

All the News That's Fit to Reprint: Do Investors React to Stale Information?

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	News data and event definition	3
3.2	Measuring staleness	3
3.3	Return, volume, and control variables	4
3.4	Individual and institutional trading data	4
4	Models and empirical specifications	5
4.1	Contemporaneous response to stale news	5
4.2	Main return-reversal regressions	5
4.3	Alternative horizons and subsamples	5
4.4	Calendar-time stale-momentum portfolios	6
4.5	Individual and institutional trading mechanism	6
4.6	Appendix model intuition	6
5	Identification and empirical design	7
5.1	What the design can identify	7
5.2	Alternative explanations addressed	7
5.3	Remaining limitations	7
6	Tables and figures	8
6.1	Tables 1 and 2: Measuring staleness and immediate market reactions	8
6.2	Table 3: Main return-reversal regressions	9
6.3	Tables 4 and 5: Horizons and subsamples	10
6.4	Tables 6 and 7: Portfolio returns and investor trading	11
6.5	Table 8: Return reversals by individual-investor activity	12

7	Short summary	13
8	Conclusion	14
8.1	Core conclusion	14
8.2	Economic intuition	14
8.3	Incremental contribution revisited	14
8.4	Implications for market efficiency	14
8.5	Implications for empirical finance	14
8.6	Limitations	15
8.7	Takeaway	15

1 Big picture

- **Question.** Do investors correctly distinguish between genuinely new information and information that is mostly a reprint or repetition of earlier news? The paper asks whether stale information still moves prices and whether those price movements subsequently reverse.
- **Setting.** The paper studies Dow Jones newswire stories about publicly traded U.S. firms from November 1996 to October 2008. A firm-specific news event is a trading day on which a qualifying story identifies the firm with a ticker code.
- **Core measurement idea.** The staleness of a story is measured using textual similarity to the previous ten news stories about the same firm. The paper computes both single-word and bigram similarity measures.
- **Main empirical pattern.** Stock returns and trading volume respond less strongly to stale news than to fresh news, consistent with stale news containing less incremental information.
- **Overreaction result.** Even though the immediate response to stale news is smaller, the return on the stale-news day negatively predicts returns over the following week. In other words, investors still react too much to redundant information.
- **Mechanism result.** Individual investors trade more aggressively in the same direction as stale news. Institutional investors do not show the same stale-news momentum trading pattern.
- **Cross-sectional mechanism.** The reversal after stale news is stronger in stocks with above-median individual-investor trading activity, which is consistent with retail price pressure helping generate the temporary mispricing.
- **Economic takeaway.** Media repetition can reduce short-run market efficiency. Investors do not fully discount information that has already been incorporated into prices, and individual trading appears to be an important channel.

2 Incremental contribution of the paper

- **From single news events to linked news sequences.** Much earlier news-event research studies isolated stories. This paper explicitly measures how similar a current story is to prior stories about the same firm, allowing the author to study interactions between news events.
- **A direct text-based measure of stale information.** Instead of classifying news manually or using coarse event categories, the paper quantifies information redundancy with Jaccard-style textual similarity scores.
- **Large-scale evidence.** The analysis covers more than 850,000 firm-news days and more than 10,000 firms, which lets the paper distinguish general market behavior from anecdotes about repeated stories.
- **Return-reversal contribution.** The paper shows that news-day returns reverse more when the news is stale, even after controlling for weekly reversal, volume-induced reversal, firm size, earnings-related news, prior volatility, illiquidity, and news intensity.
- **Portfolio contribution.** Calendar-time portfolios sorted on stale-news reactions generate negative post-news alphas, showing that the regression evidence is not just a parametric artifact.

- **Investor-type contribution.** Proprietary market-center order data identify whether individual or institutional investors trade aggressively on stale news. The paper shows the stronger stale-news trading response comes from individuals.
- **Behavioral-finance contribution.** The evidence supports a specific behavioral mechanism: some investors confuse repeated information with new information, generating temporary price pressure and later reversal.
- **Methodological contribution.** The paper demonstrates how textual similarity can be used to study information environments, news diffusion, and market microstructure questions.

3 Data and measurement

3.1 News data and event definition

- The news sample consists of Dow Jones newswire stories from November 1996 to October 2008.
- A story is linked to a firm when the Dow Jones header includes the firm’s ticker code.
- The paper keeps stories with exactly one or two U.S. ticker codes and at most three total ticker codes, so that the named firms are likely to be the main focus of the story.
- Stories with fewer than 50 words are excluded because very short stories do not provide enough text to measure similarity accurately.
- The main sample includes stories whose single-word similarity to prior stories is at least 5% or whose bigram similarity is at least 2%, increasing the chance that the story is relevant for firm valuation.
- A firm-specific news event is measured at the daily close-to-close frequency, from 4 p.m. to 4 p.m., because many news releases occur when markets are closed and the exact intraday news/information timing may not coincide.

Interpretation: The sample is designed to capture firm-specific financial news while avoiding stories in which the firm is only a minor mention. The daily event window is conservative and suitable for matching news to CRSP returns.

3.2 Measuring staleness

The paper uses textual overlap to measure whether current news repeats earlier information. For story j about firm i , let W_j^1 denote the set of unique single words and W_j^2 denote the set of unique adjacent two-word combinations, or bigrams. Pairwise single-word and bigram similarities to a previous story $j - k$ are

$$Sim1_{i,j,j-k} = \frac{\#(W_j^1 \cap W_{j-k}^1)}{\#(W_j^1 \cup W_{j-k}^1)}, \quad Sim2_{i,j,j-k} = \frac{\#(W_j^2 \cap W_{j-k}^2)}{\#(W_j^2 \cup W_{j-k}^2)}.$$

Story-level staleness is the average similarity to the previous ten stories:

$$Stale1_{ij} = \frac{1}{10} \sum_{k=1}^{10} Sim1_{i,j,j-k}, \quad Stale2_{ij} = \frac{1}{10} \sum_{k=1}^{10} Sim2_{i,j,j-k}.$$

- Single-word staleness captures repeated words and concepts.
- Bigram staleness captures repeated short phrases and is a stricter measure of textual overlap.
- Daily firm-level staleness averages the story-level measures across all qualifying stories about a firm on day t .
- The paper residualizes staleness against the number of stories, average unique words, and squared average unique words, then standardizes residual staleness within each trading day.

Interpretation: The residualized measures, $stale1_{it}$ and $stale2_{it}$, aim to capture information redundancy rather than mechanical differences in story length or news volume.

3.3 Return, volume, and control variables

- The main abnormal return, $AbRet_{it}$, is firm i 's raw return minus the CRSP value-weighted index return on day t .
- The volume measure, $AbTurn_{it}$, is log turnover on day t minus average log turnover over days $t - 5$ to $t - 1$.
- The contemporaneous volatility measure is $|AbRet_{it}|$.
- Stock-market controls include firm size, prior-week abnormal returns, prior-week idiosyncratic volatility, prior-week illiquidity, and abnormal turnover.
- News controls include the log number of messages, the log average number of unique words, earnings-related news, and prior-week abnormal newswire messages.
- All controls are standardized by trading day to make coefficients comparable across the sample.

Interpretation: The design controls for alternative sources of short-run return predictability and for mechanical features of the news environment that could be correlated with staleness.

3.4 Individual and institutional trading data

- The mechanism tests use proprietary executed market-order data from a large over-the-counter market center in Nasdaq stocks from February 2003 to December 2007.
- Orders are classified by the market center as individual or institutional based on submission and routing information.
- Individual buy-sell imbalance is

$$IndBSI_{it} = \frac{IndBuy_{it} - IndSell_{it}}{IndBuy_{it} + IndSell_{it}},$$

with an analogous institutional imbalance, $InstBSI_{it}$.

- Positive imbalances mean that the investor group is aggressively buying through market orders.
- The paper also constructs $IndAct_{it}$, the relative intensity of individual order activity compared with institutional order activity.

Interpretation: Market orders are useful for mechanism tests because they demand liquidity and can create immediate price pressure. The data allow the paper to ask which investors trade on stale news.

4 Models and empirical specifications

4.1 Contemporaneous response to stale news

The first empirical question is whether stale news looks less informative on the news day. The paper estimates daily Fama–MacBeth regressions of the form

$$DepVar_{it} = a_t + b_t StaleVar_{it} + c'_t Controls_{it} + \varepsilon_{it},$$

where $DepVar_{it}$ is either $|AbRet_{it}|$ or $AbTurn_{it}$, and $StaleVar_{it}$ is either $stale1_{it}$ or $stale2_{it}$.

Interpretation: A negative coefficient on staleness means that stale news produces smaller immediate return or volume reactions, which is what one should expect if stale stories contain less new information.

4.2 Main return-reversal regressions

The core return-predictability specification regresses post-news returns on the news-day return and its interaction with staleness:

$$Ret_{i[2,5]} = a_t + b_1 AbRet_{it} + b_2 (AbRet_{it} \times StaleVar_{it}) + c_0 StaleVar_{it} + c'_1 Controls_{it} + \varepsilon_{it}.$$

- $Ret_{i[2,5]}$ is the raw firm return from days $t + 2$ through $t + 5$.
- $AbRet_{it}$ measures the initial market reaction to the news.
- The interaction $AbRet_{it} \times StaleVar_{it}$ asks whether the initial reaction reverses more when news is stale.
- A negative coefficient on the interaction is the main stale-information result.

Interpretation: If the news-day return reverses more after stale news, the market did not fully discount the fact that the information was old.

4.3 Alternative horizons and subsamples

- The paper estimates the same framework at one-day, weekly, and two-week horizons.
- The main results are not driven by day- $t + 1$ bid-ask bounce because reversal also appears after excluding day $t + 1$.
- Subsample tests split the data by period, firm size, and earnings-related news.
- Reversal after stale news is stronger in the earlier sample period, in smaller firms, and on earnings-related news days, but the estimated interaction is negative in all major subsamples.

Interpretation: The horizon and subsample tests show that the effect is not a narrow artifact of one return window or one type of firm.

4.4 Calendar-time stale-momentum portfolios

The paper also forms portfolios using the product of news-day returns and staleness. A stale-momentum portfolio buys firms in the top decile of $AbRet_{it} \times stale_{it}$ and sells firms in the bottom decile.

- The portfolio is long stocks with positive stale-news reactions or negative fresh-news reactions.
- It is short stocks with negative stale-news reactions or positive fresh-news reactions.
- If stale-news reactions reverse, this long-short portfolio should earn negative post-news alpha.
- Alphas are estimated relative to market, size, book-to-market, and momentum factors.

Interpretation: The portfolio evidence translates the regression result into an economically intuitive trading-return format.

4.5 Individual and institutional trading mechanism

The order-flow regressions replace post-news returns with individual or institutional market-order imbalances:

$$BSI_{i[b,e]} = a_t + b_1 AbRet_{it} + b_2 (AbRet_{it} \times StaleVar_{it}) + c' Controls_{it} + \varepsilon_{it}.$$

- A positive coefficient on $AbRet_{it}$ means traders buy after positive news and sell after negative news.
- A positive coefficient on $AbRet_{it} \times StaleVar_{it}$ means this aggressive same-direction trading is stronger when news is stale.
- The key result is that the stale-news trading interaction is positive and significant for individuals on day t , but not for institutions.

Interpretation: The mechanism evidence links stale-news overreaction to individual-investor trading pressure rather than institutional trading.

4.6 Appendix model intuition

The appendix model contains rational investors and boundedly rational investors. A first signal s_1 is released, and then a second signal $s_1 + s_2$ is released. Rational investors separate old information s_1 from new information s_2 . Boundedly rational investors partially treat the old component as new information.

The market-clearing prices in the model are

$$p_2 = (1 + mk)s_1 + s_2, \quad p_1 = [1 + m(1 - m)k]s_1,$$

where m is the fraction of boundedly rational investors and k measures the degree to which stale information is treated as new.

Interpretation: When more stale information is mistakenly treated as new, prices overreact and subsequent returns reverse. Reversal should be larger when recent stale information is abundant and when the fraction of investors prone to this error is larger.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies whether returns reverse more after news that is more textually similar to recent firm-specific news.
- The design is not a randomized experiment. It does not randomly assign stale versus fresh news.
- The key identifying comparison is cross-sectional: among firms with news on the same day, are reactions to more stale news more likely to reverse?
- Daily standardization and Fama–MacBeth estimation reduce concerns that the result is driven by aggregate market conditions.

Interpretation: The paper is strongest as evidence of a predictable price pattern associated with measured information redundancy. The mechanism evidence from order flow makes the behavioral interpretation more convincing.

5.2 Alternative explanations addressed

- **Low information content.** Stale news produces smaller same-day responses, but the reversal result remains even within earnings-related stories, so staleness is not simply irrelevant news.
- **Generic short-term reversal.** The return regressions control for prior-week reversal and day- t returns interacted with several known reversal predictors.
- **Volume-induced reversal.** Abnormal turnover and interactions with turnover are included in richer specifications.
- **Bid-ask bounce.** The main post-news return window begins at day $t + 2$, excluding the adjacent day.
- **Risk exposure.** Calendar-time portfolios control for standard return factors.
- **Institutional trading.** The order-flow evidence shows that the stale-news same-direction trading response is concentrated among individuals, not institutions.

5.3 Remaining limitations

- Textual similarity is an imperfect proxy for economic redundancy. Similar words can sometimes describe materially new facts, and different words can sometimes describe the same fact.

- The staleness measure is based on similarity to prior Dow Jones stories, not all possible information sources such as analyst reports, press releases, or other media.
- The order-flow analysis covers Nasdaq stocks and the 2003–2007 market-center sample, not the full 1996–2008 news sample.
- Some return predictability may be hard to implement because real-time individual/institutional trading intensity is not generally observable to all traders.

6 Tables and figures

6.1 Tables 1 and 2: Measuring staleness and immediate market reactions

Read:

- Table 1 reports the distribution of raw and residual single-word and bigram staleness measures.
- Raw single-word staleness has a mean of 0.117, while raw bigram staleness has a mean of 0.077. Residual measures are standardized to mean zero and standard deviation one.
- Table 1 also shows that the residual staleness measures have low correlations with many standard firm and news characteristics.
- Table 2 shows that stale news is associated with lower contemporaneous absolute returns and lower abnormal turnover.
- The negative staleness-return-volatility relation is much stronger for small firms than for big firms, which the paper interprets as possible evidence that staleness is harder to measure for information-rich large firms.

Economic message: These tables validate the staleness measure. Stale stories look less informative on the event day, and the measure is not just a repackaging of news volume, story length, earnings news, size, turnover, or volatility.

Table 1
Cross-sectional distributions and correlations for the two staleness measures

Panel A: Distributions of the Raw and Residual Staleness Measures				
	$Stale1_u$	$Stale2_u$	$stale1_u$	$stale2_u$
Mean	0.117	0.077	0.000	0.000
Standard Deviation	0.068	0.064	1.000	1.000
5th Percentile	0.058	0.028	-1.033	-0.969
10th Percentile	0.063	0.031	-0.882	-0.827
25th Percentile	0.075	0.039	-0.606	-0.580
50th Percentile	0.097	0.056	-0.230	-0.236
75th Percentile	0.133	0.090	0.306	0.284
90th Percentile	0.194	0.150	1.110	1.118
95th Percentile	0.251	0.206	1.867	1.920
Panel B: Correlations of Firm and News Characteristics with Staleness Measures				
	$Stale1_u$	$Stale2_u$	$stale1_u$	$stale2_u$
Msg_u	0.117**	0.044**	0.000	-0.006**
$Words_u$	-0.213**	-0.130**	0.000	0.002**
$(Words_u)^2$	-0.006	0.000	0.000	0.001**
$Earn_u$	-0.089**	-0.120**	-0.057**	-0.089**
$AbRet_{it}$	-0.107**	-0.007**	-0.003**	-0.002
$ AbRet_{it} $	-0.019**	-0.055**	-0.032**	-0.054**
$AbTurn_u$	-0.039**	-0.090**	-0.058**	-0.089**
$MktCap_{i,t-1}$	0.023**	0.055**	0.004	0.040**
$Msg[-5,-1]$	0.109**	0.120**	0.117**	0.116**
$IdVolat[-5,-1]$	0.002	-0.005	0.016**	0.003
$Illiq[-5,-1]$	-0.017**	-0.052**	-0.006*	-0.041**

Panel A in Table 1 summarizes the cross-sectional statistical properties of the single word, and the bigram raw and residual textual staleness measures $Stale1_{it}$ and $Stale2_{it}$, and $stale1_{it}$ and $stale2_{it}$. For each measure, it shows the average of the daily cross-sectional mean, standard deviation, and the 5th, 25th, 50th, 75th, and 95th percentiles. Both residual staleness measures are orthogonalized with respect to three news story characteristics and standardized by day, as explained in the text. Panel B presents the daily average of the correlations of each of the four staleness measures with several control variables. Some control variables measure characteristics of news stories, including newswire messages (Msg_{it}), prior-week abnormal newswire messages ($Msg[-5,-1]$), words in the news ($Words_{it}$), squared words ($(Words_{it})^2$), and earnings-related news ($Earn_{it}$). Other control variables measure firm characteristics, including firm size ($MktCap_{i,t-1}$), abnormal turnover ($AbTurn_{it}$), prior-week idiosyncratic volatility ($IdVolat[-5,-1]$), prior-week illiquidity ($Illiq[-5,-1]$), and cumulative abnormal returns in the past week ($AbRet_{it}[-5,-1]$). See Section 1 for further details. In Panel B, the symbols **, *, and + denote statistical significance at the 1%, 5%, and 10% levels, respectively. Significance is based on Newey-West (1987) standard errors that are robust to heteroskedasticity and five days of autocorrelation.

Table 2
Relating absolute returns and trading volume to news staleness

Dependent Variable	$ AbRet_{it} $	$ AbRet_{it} $	$ AbRet_{it} $	$ AbRet_{it} $	$AbTurn_{it}$	$AbTurn_{it}$
Firms Included	All	All	Small	Big	All	All
$stale1_u$	-0.104 (0.007)		-0.203 (0.012)	-0.036 (0.005)	-0.038 (0.002)	
$stale2_u$		-0.143 (0.008)				-0.048 (0.002)
Msg_u	1.111 (0.015)	1.046 (0.016)	1.740 (0.030)	0.536 (0.011)	0.227 (0.003)	0.217 (0.003)
$Msg[-5,-1]$	-0.168 (0.007)	-0.150 (0.007)	-0.188 (0.010)	-0.092 (0.005)	-0.039 (0.001)	-0.037 (0.001)
$Words_u$	0.108 (0.007)	0.085 (0.007)	0.396 (0.016)	-0.005 (0.005)	0.000 (0.002)	-0.005 (0.002)
$Earn_u$	0.092 (0.007)	0.080 (0.007)	0.040 (0.011)	0.135 (0.006)	0.056 (0.001)	0.050 (0.001)
$MktCap_{i,t-1}$	-1.221 (0.018)	-1.152 (0.018)	-1.767 (0.042)	-0.789 (0.015)	-0.160 (0.003)	-0.152 (0.003)
$IdVolat[-5,-1]$	0.836 (0.021)	0.776 (0.020)	0.828 (0.023)	0.805 (0.028)	-0.014 (0.002)	-0.018 (0.002)
$Illiq[-5,-1]$	-0.244 (0.011)	-0.216 (0.012)	-0.387 (0.016)	-1.383 (0.122)	0.060 (0.003)	0.058 (0.003)
$AbRet[-5,-1]$	-0.185 (0.013)	-0.159 (0.014)	-0.197 (0.015)	-0.151 (0.013)	-0.018 (0.002)	-0.017 (0.002)
Average Value	2.894 (0.043)	2.733 (0.043)	2.614 (0.050)	1.935 (0.070)	0.262 (0.006)	0.239 (0.007)
Trading Days	2853	2652	2367	2238	2853	2652
Obs per Day	333	305	197	178	333	305
R^2	20.7%	20.9%	20.2%	21.2%	18.8%	18.6%

Table 2 displays results from daily cross-sectional Fama-MacBeth (1973) regressions of either absolute abnormal returns ($|AbRet_{it}|$) or abnormal turnover ($AbTurn_{it}$) during day t on news and control variables known by day t . The table reports the time-series average of the coefficients, R^2 statistics, and average values of the dependent variable. The single-word and bigram measures of textual staleness are $stale1_{it}$ and $stale2_{it}$. Control variables include newswire messages (Msg_{it}), prior-week abnormal newswire messages ($Msg[-5,-1]$), words in the news ($Words_{it}$), earnings-related news ($Earn_{it}$), firm size ($MktCap_{i,t-1}$), abnormal turnover ($AbTurn_{it}$), prior-week idiosyncratic volatility ($IdVolat[-5,-1]$), prior-week illiquidity ($Illiq[-5,-1]$), and cumulative abnormal returns in the past week ($AbRet_{it}[-5,-1]$). See the text for further details. The regression intercept is the average value of the dependent variable on day t . Columns three and four in the table separately examine subsamples of below-median and above-median firm size. I standardize all independent variables by day so that the coefficient units are basis points per standard-deviation increase in the independent variables. Newey-West (1987) standard errors robust to heteroskedasticity and five days of autocorrelation appear in parentheses. All coefficients are statistically significant at the 1% level.

(a) Table 1: Staleness distributions and correlations

(b) Table 2: Absolute returns and turnover around stale news

6.2 Table 3: Main return-reversal regressions

Read:

- Table 3 predicts post-news returns from days $t + 2$ through $t + 5$.
- The key coefficients are on $AbRet_{it} \times stale1_{it}$ and $AbRet_{it} \times stale2_{it}$.
- These coefficients are negative and statistically significant across specifications with no controls, main controls, and all controls.
- The negative interaction means that a stronger news-day return is more likely to reverse when the news is stale.
- The result survives controls for earnings-related news, news intensity, size, prior-week returns, abnormal turnover, prior volatility, illiquidity, story length, and prior news coverage.

Economic message: Table 3 is the central return-predictability result. Investors partly recognize that stale news is less informative, but not enough: the remaining stale-news price response is still too strong and later reverses.

Table 3
Predicting return reversals after news using staleness and control variables

Dependent Variable	$Ret_t[2,5]$	$Ret_t[2,5]$	$Ret_t[2,5]$	$Ret_t[2,5]$	$Ret_t[2,5]$	$Ret_t[2,5]$
Controls Included	None	None	Main	Main	All	All
$AbRet_{it}$	0.015 (0.019)	-0.003 (0.020)	-0.049* (0.022)	-0.076** (0.024)	-0.053* (0.024)	-0.093** (0.026)
$AbRet_{it} * stale1_{it}$	-0.057** (0.016)		-0.059** (0.016)		-0.066** (0.017)	
$stale1_{it}$	-0.024* (0.014)		-0.007 (0.012)		-0.007 (0.012)	
$AbRet_{it} * stale2_{it}$		-0.053** (0.016)		-0.046** (0.017)		-0.047** (0.018)
$stale2_{it}$		-0.029* (0.016)		-0.025* (0.014)		-0.021 (0.014)
$AbRet_{it} * Msg_{it}$			0.057** (0.014)	0.069** (0.015)	0.023 (0.016)	0.027 (0.018)
Msg_{it}			-0.033** (0.012)	-0.018 (0.013)	-0.030* (0.012)	-0.017 (0.013)
$AbRet_{it} * Earn_{it}$			0.032* (0.013)	0.021 (0.015)	0.032* (0.014)	0.017 (0.017)
$Earn_{it}$			0.001 (0.011)	-0.002 (0.011)	0.001 (0.011)	-0.002 (0.011)
$AbRet_{it} * MktCap_{it-1}$			-0.065** (0.018)	-0.063** (0.020)	-0.122** (0.027)	-0.114** (0.028)
$MktCap_{it-1}$			-0.115** (0.029)	-0.103** (0.030)	-0.116** (0.029)	-0.108** (0.030)
$AbRet_{it}[-5,-1]$			-0.060** (0.022)	-0.060** (0.023)	-0.059** (0.023)	-0.061** (0.023)
$AbTurn_{it}$			0.060** (0.011)	0.054** (0.011)	0.055** (0.011)	0.053** (0.011)
$IdVolat_{it}[-5,-1]$			-0.067* (0.030)	-0.044 (0.031)	-0.060* (0.033)	-0.044 (0.033)
$Illiq_{it}[-5,-1]$			-0.156** (0.023)	-0.141** (0.023)	-0.157** (0.023)	-0.144** (0.024)
$Words_{it}$			-0.030* (0.012)	-0.031** (0.012)	-0.028* (0.012)	-0.029* (0.012)
$Msg_{it}[-5,-1]$			0.006 (0.010)	0.013 (0.010)	0.005 (0.010)	0.014 (0.010)
$AbRet_{it} * AbTurn_{it}$					0.038** (0.014)	0.045** (0.015)
$AbRet_{it} * IdVolat_{it}[-5,-1]$					-0.092** (0.015)	-0.076** (0.016)
$AbRet_{it} * Illiq_{it}[-5,-1]$					-0.045* (0.018)	-0.051** (0.020)

continued

Table 3
Continued

$AbRet_{it} * Words_{it}$					0.006 (0.016)	-0.007 (0.017)
$AbRet_{it} * Msg_{it}[-5,-1]$					-0.007 (0.014)	-0.011 (0.016)
Average $Ret_t[2,5]$	0.101 (0.086)	0.108 (0.086)	0.112 (0.087)	0.114 (0.088)	0.107 (0.087)	0.109 (0.088)
Trading Days	2863	2661	2853	2652	2853	2652
Observations per Day	337	309	333	305	333	305
R^2	2.7%	2.9%	12.3%	13.0%	15.8%	16.8%

Table 3 presents results from daily cross-sectional Fama-MacBeth (1973) regressions of raw post-news firm returns from days $t+2$ through $t+5$ on news-related variables and other control variables known by day t . The table reports the time-series average of the coefficients, return from days $t+2$ through $t+5$, and R^2 statistics. The independent variable for the firm's abnormal return on day t ($AbRet_{it}$) measures return reversal. The interaction terms with $AbRet_{it}$ allow reversal to depend on the two textual staleness measures ($stale1_{it}$ and $stale2_{it}$), newswire messages (Msg_{it}), earnings-related news ($Earn_{it}$), firm size ($MktCap_{it-1}$), abnormal turnover ($AbTurn_{it}$), prior-week idiosyncratic volatility ($IdVolat_{it}[-5,-1]$), prior-week illiquidity ($Illiq_{it}[-5,-1]$), words in the news ($Words_{it}$), and prior-week abnormal newswire messages ($Msg_{it}[-5,-1]$). Each regression controls for the direct effects of all included interaction term variables. The other control variable is the firm's cumulative abnormal return in the past week ($AbRet_{it}[-5,-1]$). See text for variable construction details. I standardize all independent variables by day so that the coefficient units are basis points per standard-deviation increase in the independent variables. Newey-West (1987) standard errors that are robust to heteroskedasticity and five days of autocorrelation appear in parentheses. The symbols **, *, and + denote statistical significance at the 1%, 5%, and 10% levels, respectively.

(a) Table 3, part 1: Main reversal coefficients

(b) Table 3, part 2: Continued controls and table notes

6.3 Tables 4 and 5: Horizons and subsamples

Read:

- Table 4 changes the post-news return horizon. The stale-news reversal appears over days $t+2$ to $t+10$ and is not confined to day $t+1$.
- The one-day estimates are smaller and noisier, but the signs are broadly consistent with reversal.
- Table 5 splits the sample by time period, firm size, and earnings news.
- The stale-news interaction is negative in the 1996–2001 and 2002–2008 subsamples, in small and big firms, and in both earnings and non-earnings news subsamples.
- The effect is strongest in the earlier period, in smaller firms, and in earnings-related news.

Economic message: The stale-news reversal is not an artifact of one horizon or one sample slice. It is a broad pattern, though it is economically strongest where behavioral trading and information frictions are likely to matter more.

Table 4
Predicting return reversals at different time horizons using staleness

Dependent Variable	$Ret[1,1]$	$Ret[1,1]$	$Ret[2,2]$	$Ret[2,2]$	$Ret[2,10]$	$Ret[2,10]$
Controls Included	Main	Main	Main	Main	Main	Main
$AbRet_{it}$	0.041** (0.013)	0.025* (0.013)	-0.043** (0.012)	-0.049** (0.012)	-0.042 (0.032)	-0.078* (0.034)
$AbRet_{it} * staleI_{it}$	-0.012 (0.010)		-0.015* (0.008)		-0.069** (0.022)	
$staleI_{it}$	-0.009* (0.005)		-0.008 (0.005)		-0.031 (0.021)	
$AbRet_{it} * stale2_{it}$		-0.010 (0.011)		-0.014 (0.009)		-0.052* (0.024)
$stale2_{it}$		-0.010* (0.005)		-0.010* (0.005)		-0.048* (0.025)
$AbRet_{it} * Msg_{it}$	0.038** (0.009)	0.048** (0.009)	0.023** (0.007)	0.032** (0.007)	0.088** (0.018)	0.101** (0.019)
Msg_{it}	-0.029** (0.005)	-0.031** (0.006)	-0.021** (0.005)	-0.014** (0.005)	-0.051** (0.017)	-0.032 (0.019)
$AbRet_{it} * Earn_{it}$	0.043** (0.009)	0.050** (0.010)	0.015* (0.007)	0.012 (0.008)	0.048** (0.018)	0.043* (0.020)
$Earn_{it}$	0.010* (0.005)	0.003 (0.005)	-0.011** (0.004)	-0.008 (0.005)	0.032* (0.018)	0.039* (0.018)
$AbRet_{it} * MktCap_{it-1}$	-0.004 (0.011)	-0.010 (0.011)	-0.021* (0.010)	-0.018* (0.010)	-0.066** (0.026)	-0.057 (0.028)
$MktCap_{it-1}$	-0.017* (0.010)	-0.013 (0.010)	-0.013 (0.009)	-0.006 (0.010)	-0.182** (0.052)	-0.163** (0.052)
$AbRet_{it}[-5,-1]$	-0.087** (0.010)	-0.083** (0.011)	-0.039** (0.009)	-0.036** (0.009)	-0.022 (0.039)	-0.017 (0.040)
Average Return	0.042* (0.025)	0.039 (0.025)	0.026 (0.024)	0.022 (0.025)	0.254 (0.159)	0.265 (0.162)
Trading Days	2853	2652	2853	2652	2853	2652
Obs per Day	333	305	333	305	333	305
R ²	13.6%	14.3%	12.2%	13.0%	11.9%	12.6%

Table 4 presents daily cross-sectional Fama-MacBeth (1973) regressions of raw firm returns from days $t+1$ through $t+5$ on news variables and control variables known by day t . The three $[b, c]$ intervals in the columns are $[1, 1]$, $[2, 2]$, and $[2, 10]$. The table reports the time-series average of the regression coefficients, returns on days $[b, c]$, and R^2 statistics. The coefficient on the firm's abnormal return on day t ($AbRet_{it}$) measures return reversal. Interaction terms with $AbRet_{it}$ allow reversal to depend on the two textual staleness measures ($staleI_{it}$ and $stale2_{it}$), newswire messages (Msg_{it}), earnings news ($Earn_{it}$), and firm size ($MktCap_{it-1}$). Other controls include the direct effects of these variables, weekly returns ($AbRet_{it}[-5,-1]$), turnover ($AbTurn_{it}$), weekly volatility ($IdVolat_{it}[-5,-1]$), weekly illiquidity ($Illiq_{it}[-5,-1]$), words in the news ($Words_{it}$), and weekly newswire messages ($Msg_{it}[-5,-1]$). The coefficient units are basis points per standard deviation of the independent variables. Newey-West (1987) standard errors robust to heteroskedasticity and five days of autocorrelation appear in parentheses. The symbols **, *, and + denote statistical significance at the 1%, 5%, and 10% levels, respectively.

(a) Table 4: Return-reversal horizons

Table 5
Predicting return reversals using staleness in different subsamples

Dependent Variable	$Ret[2,5]$	$Ret[2,5]$	$Ret[2,5]$	$Ret[2,5]$	$Ret[2,5]$	$Ret[2,5]$
Subsample	96 to 01	02 to 08	Small	Big	Earnings	No Earnings
$AbRet_{it}$	-0.067* (0.037)	-0.032 (0.024)	-0.008 (0.039)	-0.010 (0.044)	0.026 (0.042)	-0.112** (0.027)
$AbRet_{it} * staleI_{it}$	-0.092** (0.026)	-0.027 (0.018)	-0.082** (0.020)	-0.038 (0.027)	-0.102* (0.043)	-0.049* (0.022)
$staleI_{it}$	-0.028 (0.022)	0.014 (0.010)	-0.007 (0.017)	-0.013 (0.014)	0.035 (0.032)	-0.013 (0.013)
$AbRet_{it} * Msg_{it}$	0.088** (0.024)	0.026 (0.014)	0.046** (0.016)	0.120** (0.024)	0.035 (0.025)	0.043* (0.022)
Msg_{it}	-0.053** (0.020)	-0.013 (0.012)	-0.086** (0.020)	-0.026 (0.014)	-0.014 (0.024)	-0.040** (0.015)
$AbRet_{it} * Earn_{it}$	0.030 (0.024)	0.033** (0.013)	0.008 (0.017)	0.046* (0.027)		
$Earn_{it}$	0.002 (0.020)	-0.001 (0.009)	0.029 (0.015)	-0.028 (0.013)		
$AbRet_{it} * MktCap_{it-1}$	-0.048 (0.032)	-0.083** (0.017)	-0.013 (0.035)	-0.208** (0.044)	-0.046 (0.029)	-0.095** (0.023)
$MktCap_{it-1}$	-0.139** (0.052)	-0.090** (0.028)	-0.438** (0.062)	-0.037 (0.034)	-0.105 (0.051)	-0.091** (0.033)
$AbRet_{it}[-5,-1]$	-0.101* (0.040)	-0.020 (0.020)	-0.054* (0.024)	-0.056 (0.035)	-0.038 (0.038)	-0.087** (0.026)
Average $Ret[2,5]$	0.134 (0.137)	0.091 (0.106)	-0.159 (0.100)	-0.340* (0.159)	0.097 (0.117)	0.112 (0.092)
Trading Days	1418	1435	2367	2238	925	2504
Obs per Day	273	391	197	178	165	257
R ²	13.8%	10.8%	14.1%	17.7%	15.8%	12.9%

Table 5 presents daily cross-sectional Fama-MacBeth (1973) regressions of raw firm returns from days $t+2$ through $t+5$ on news variables and control variables measured prior to day t . The six columns show different subsamples: years 1996 to 2001 and 2002 to 2008; above-average firm size ($MktCap_{it-1}$) and below-average firm size; news days with earnings-related words as measured by $Earn_{it}$, and news days without earnings news. The table reports the time-series average of the regression coefficients, returns on days $t+2$ through $t+5$, and R^2 statistics. The coefficient on the firm's abnormal return on day t ($AbRet_{it}$) measures return reversal. Interaction terms with $AbRet_{it}$ allow reversal to depend on the single-word staleness measure ($staleI_{it}$), newswire messages (Msg_{it}), earnings news ($Earn_{it}$), and firm size ($MktCap_{it-1}$). Other controls include the direct effects of these variables, weekly returns ($AbRet_{it}[-5,-1]$), turnover ($AbTurn_{it}$), weekly volatility ($IdVolat_{it}[-5,-1]$), weekly illiquidity ($Illiq_{it}[-5,-1]$), words in the news ($Words_{it}$), and weekly newswire messages ($Msg_{it}[-5,-1]$). The coefficient units are basis points per standard deviation of the independent variables. Newey-West (1987) standard errors robust to heteroskedasticity and five days of autocorrelation appear in parentheses. The symbols **, *, and + denote statistical significance at the 1%, 5%, and 10% levels, respectively.

(b) Table 5: Return-reversal subsamples

6.4 Tables 6 and 7: Portfolio returns and investor trading

Read:

- Table 6 forms calendar-time portfolios from the interaction of news-day returns and staleness.
- The stale-momentum portfolios earn negative post-news alphas, meaning that stale-news reactions reverse in portfolio returns.
- The equal-weighted cumulative alpha over days $t+1$ to $t+5$ is about -26 to -27 basis points, depending on the staleness measure.
- Table 7 turns to market-order imbalances by investor type.
- Individual investors trade aggressively in the same direction as stale news on day t ; the corresponding institutional coefficients are not positive in the same way.
- Individual same-direction stale-news trading fades after the event day, consistent with temporary pressure that later reverses.

Economic message: The portfolio evidence shows that stale-news reversal is economically meaningful, and the order-flow evidence points to individual investors as the group most likely to mistake repeated information for new information.

Table 6

Returns of calendar-time portfolios formed on reactions to stale news

Panel A: Equal-weighted Returns, Formed on Top minus Bottom Decile of $AbRet_{it} \times stale1_{it}$						
Holding Period	[0,0]	[1,1]	[2,2]	[2,5]	[1,5]	[1,10]
<i>MKT</i>	-0.079 (0.057)	0.001 (0.026)	-0.019 (0.023)	0.007 (0.014)	0.011 (0.042)	-0.002 (0.011)
<i>SMB</i>	-0.333** (0.118)	-0.020 (0.056)	0.017 (0.040)	0.027 (0.025)	0.020 (0.023)	-0.002 (0.015)
<i>HML</i>	-0.156 (0.125)	0.031 (0.054)	0.027 (0.045)	0.030 (0.026)	0.034 (0.026)	0.026 (0.021)
<i>UMD</i>	-0.036 (0.081)	0.037 (0.034)	0.027 (0.030)	0.033* (0.018)	0.043** (0.016)	0.034** (0.013)
<i>Daily Alpha</i>	-4.133** (0.100)	-0.089** (0.026)	-0.059** (0.021)	-0.043** (0.011)	-0.052** (0.011)	-0.030** (0.008)
<i>Cumulative Alpha</i>	-4.133** (0.100)	-0.089** (0.026)	-0.059** (0.021)	-0.175** (0.045)	-0.262** (0.055)	-0.301** (0.085)
Trading Days	2863	2861	2859	2947	2956	2980
R^2	0.3%	0.1%	0.1%	0.2%	0.3%	0.3%
Panel B: Equal-weighted Returns, Formed on Top minus Bottom Decile of $AbRet_{it} \times stale2_{it}$						
Holding Period	[0,0]	[1,1]	[2,2]	[2,5]	[1,5]	[1,10]
<i>Cumulative Alpha</i>	-5.219** (0.103)	-0.140** (0.027)	-0.033 (0.024)	-0.129** (0.050)	-0.270** (0.065)	-0.210* (0.107)
Trading Days	2662	2661	2659	2830	2851	2911
R^2	0.2%	0.1%	0.3%	0.2%	0.1%	0.2%
Panel C: Value-weighted Returns, Formed on Top minus Bottom Decile of $AbRet_{it} \times stale1_{it}$						
Holding Period	[0,0]	[1,1]	[2,2]	[2,5]	[1,5]	[1,10]
<i>Cumulative Alpha</i>	-0.748** (0.092)	-0.057* (0.032)	-0.110** (0.033)	-0.069 (0.076)	-0.148* (0.086)	-0.186* (0.116)
R^2	0.2%	0.2%	0.1%	0.0%	0.1%	0.1%
Panel D: Value-weighted Returns, Formed on Top minus Bottom Decile of $AbRet_{it} \times stale2_{it}$						
Holding Period	[0,0]	[1,1]	[2,2]	[2,5]	[1,5]	[1,10]
<i>Cumulative Alpha</i>	-0.876** (0.089)	-0.062* (0.033)	-0.125** (0.032)	-0.153* (0.070)	-0.168** (0.064)	-0.144 (0.128)
R^2	0.2%	0.1%	0.2%	0.5%	0.4%	0.2%

Table 6 displays the risk factor loadings and daily alphas of stale momentum portfolios. These portfolios are formed each day by buying firms in the top decile of either $AbRet_{it} \times stale1_{it}$ or $AbRet_{it} \times stale2_{it}$, and selling firms in the bottom decile of $AbRet_{it} \times stale1_{it}$ or $AbRet_{it} \times stale2_{it}$. The four panels show the daily equal- and value-weighted returns for the reversal portfolios using the two (single-word or bigram) staleness measures. The columns show reversal portfolios held for different time horizons after news events on day t : days [0,0], [1,1], [2,2], [2,5], [1,5], and [1,10]. Each daily alpha is the intercept from a time-series regression of portfolio returns on four factors: the market (MKT), size (SMB), book-to-market (HML), and momentum (UMD) factors available on Kenneth French's website. Cumulative alphas equal daily alphas times days in the holding period. Newey and West (1987) standard errors robust to heteroskedasticity and five days of autocorrelation appear in parentheses below the coefficients. The symbols **, *, and + denote statistical significance at the 1%, 5%, and 10% levels, respectively.

(a) Table 6: Calendar-time stale-momentum portfolios

Table 7

Individual and institutional market order imbalances after news

Dependent Variable	<i>IndBSI</i>	<i>IndBSI</i>	<i>IndBSI</i>	<i>IndBSI</i>	<i>InstImb</i>	<i>InstBSI</i>	<i>InstBSI</i>	<i>InstBSI</i>
Time Horizon	[0,0]	[0,0]	[1,1]	[2,5]	[0,0]	[0,0]	[1,1]	[2,5]
Controls Included	Main	Main	Main	Main	Main	Main	Main	Main
<i>AbRet_{it}</i>	4.137** (0.219)	3.825** (0.215)	1.266** (0.199)	-0.023 (0.126)	1.339** (0.179)	1.078** (0.174)	0.459* (0.172)	0.021 (0.090)
<i>AbRet_{it} × stale1_{it}</i>	0.900** (0.196)	0.212 (0.182)	-0.134 (0.107)	-0.051 (0.138)			-0.195 (0.132)	0.077 (0.071)
<i>stale1_{it}</i>	0.176 (0.145)	0.195* (0.120)	0.479** (0.087)	0.335** (0.086)			0.149* (0.080)	0.239** (0.051)
<i>AbRet_{it} × stale2_{it}</i>		1.045** (0.227)				0.049 (0.143)		
<i>stale2_{it}</i>		0.224 (0.153)				0.341** (0.090)		
<i>AbRet_{it} × Msg_{it}</i>	-3.029** (0.137)	-2.651** (0.144)	-1.255** (0.114)	-0.372** (0.073)	-0.566** (0.113)	-0.463** (0.114)	-0.279** (0.104)	0.045 (0.049)
<i>Msg_{it}</i>	-0.406** (0.146)	-0.046 (0.148)	-0.971** (0.134)	-0.725** (0.087)	-0.320** (0.106)	-0.288** (0.106)	-0.364** (0.098)	-0.161** (0.054)
<i>AbRet_{it} × Earn_{it}</i>	-0.372** (0.126)	-0.407** (0.145)	-0.115 (0.140)	0.173* (0.084)	-0.072 (0.137)	-0.119 (0.140)	0.036 (0.112)	0.125* (0.056)
<i>Earn_{it}</i>	0.215 (0.137)	0.053 (0.150)	0.225* (0.129)	0.316** (0.080)	0.026 (0.086)	0.065 (0.098)	-0.074 (0.085)	0.010 (0.046)
<i>AbRet_{it} × MktCap_{it-1}</i>	-1.330** (0.174)	-1.322** (0.184)	-0.217 (0.175)	-0.130 (0.103)	0.702** (0.132)	0.444** (0.144)	0.278* (0.145)	0.044 (0.071)
<i>MktCap_{it-1}</i>	-2.843** (0.346)	-3.231** (0.368)	-2.084** (0.368)	-2.262** (0.320)	-1.327** (0.271)	-1.431** (0.267)	-1.385** (0.248)	-1.636** (0.217)
<i>AbRet_{it} × [-5, -1]</i>	-0.441** (0.141)	-0.307* (0.148)	-0.774** (0.133)	-0.443** (0.083)	0.029 (0.100)	0.048 (0.110)	0.065 (0.091)	0.024 (0.059)
<i>Average Imbalance</i>	-7.379** (0.480)	-8.118** (0.454)	-7.343** (0.394)	-7.611** (0.376)	0.087 (0.228)	-0.138 (0.229)	-0.005 (0.209)	0.049 (0.178)
Trading Days	1148	1084	1148	1148	1148	1084	1148	1148
Obs per Day	301	275	301	301	301	275	301	301
R^2	7.4%	7.9%	7.3%	8.1%	7.9%	8.3%	7.5%	7.9%

Table 7 presents daily cross-sectional Fama-MacBeth (1973) regressions of individual (*IndBSI*) and institutional market order imbalances (*InstBSI*) from days $t + b$ through $t + e$ on news and control variables. The three columns show $[b, e]$ intervals of [0,0], [1,1], and [2,5]. The table reports the time-series average of the regression coefficients, imbalances on days $[b, e]$, and R^2 statistics. The coefficient on the firm's abnormal return on day t ($AbRet_{it}$) measures aggressive momentum or contrarian trading. Interaction terms with $AbRet_{it}$ allow momentum trading to depend on the two textual staleness measures ($stale1_{it}$ and $stale2_{it}$), newswire messages (Msg_{it}), earnings news ($Earn_{it}$), and firm size ($MktCap_{it-1}$). The complete set of independent variables is the same as in Tables 4 and 5—see those tables for details. The coefficient units are basis points per standard deviation of the independent variables. Newey-West (1987) standard errors robust to heteroskedasticity and five days of autocorrelation appear in parentheses. The symbols **, *, and + denote statistical significance at the 1%, 5%, and 10% levels, respectively.

(b) Table 7: Individual and institutional order imbalances

6.5 Table 8: Return reversals by individual-investor activity

Read:

- Table 8 splits firms by above- and below-median individual-investor trading activity.
- The coefficients on $AbRet_{it} \times stale1_{it}$ and $AbRet_{it} \times stale2_{it}$ are negative and significant when individual activity is high.
- The corresponding coefficients are near zero when individual activity is low.
- This table is restricted to the 2003–2007 order-flow sample, so it has fewer observations than the main return-reversal tests.

Economic message: Table 8 closes the mechanism loop. Stale-news reversal is concentrated precisely where individual-investor activity is high, supporting the interpretation that individual trading pressure generates temporary mispricing.

Table 8
Predicting return reversals using staleness and individual investor activity

Dependent Variable	$Ret_{[2,5]}$	$Ret_{[2,5]}$	$Ret_{[2,5]}$	$Ret_{[2,5]}$
Individual Activity	High	Low	High	Low
Controls Included	Main	Main	Main	Main
$AbRet_{it}$	-0.044 (0.030)	-0.035 (0.034)	-0.069* (0.032)	-0.050 (0.035)
$AbRet_{it} * stale1_{it}$	-0.063* (0.026)	0.002 (0.037)		
$stale1_{it}$	0.026 (0.018)	-0.004 (0.014)		
$AbRet_{it} * stale2_{it}$			-0.091** (0.027)	0.020 (0.040)
$stale2_{it}$			0.033 (0.018)	-0.013 (0.015)
$AbRet_{it} * Msg_{it}$	0.011 (0.018)	0.035 (0.025)	0.016 (0.020)	0.032 (0.026)
Msg_{it}	-0.013 (0.019)	-0.018 (0.012)	-0.002 (0.019)	-0.016 (0.013)
$AbRet_{it} * Earn_{it}$	0.021 (0.020)	0.046* (0.027)	0.018 (0.021)	0.054* (0.027)
$Earn_{it}$	-0.002 (0.018)	-0.002 (0.013)	-0.012 (0.020)	-0.004 (0.014)
$AbRet_{it} * MktCap_{i,t-1}$	-0.105** (0.028)	-0.082** (0.027)	-0.113** (0.027)	-0.079** (0.030)
$MktCap_{i,t-1}$	-0.078* (0.041)	-0.064* (0.028)	-0.089** (0.042)	-0.066* (0.030)
$AbRet_{it} [-5, -1]$	-0.001 (0.024)	-0.021 (0.028)	0.003 (0.025)	-0.031 (0.030)
Average Imbalance	0.142 (0.107)	0.186* (0.096)	0.120 (0.107)	0.193* (0.098)
Trading Days	967	970	912	919
Obs per Day	165	165	149	149
R^2	15.2%	18.0%	16.3%	19.7%

Table 8 presents daily cross-sectional Fama-MacBeth (1973) regressions of raw firm returns from days $t + 2$ through $t + 5$ on news variables and control variables known by day t . The columns show subsamples with above-median and below-median individual investor trading activity ($IndAct_{it}$). The table reports the time-series average of the regression coefficients, returns on days $t + 2$ through $t + 5$, and R^2 statistics. The coefficient on the firm's abnormal return on day t ($AbRet_{it}$) measures return reversal. Interaction terms with $AbRet_{it}$ allow reversal to depend on the two textual staleness measures ($stale1_{it}$ and $stale2_{it}$), newswire messages (Msg_{it}), earnings news ($Earn_{it}$), and firm size ($MktCap_{i,t-1}$). The complete set of independent variables is the same as in Tables 4 and 5—see those tables for details. The coefficient units are basis points per standard deviation of the independent variables. Newey-West (1987) standard errors robust to heteroskedasticity and five days of autocorrelation appear in parentheses. The symbols **, *, and + denote statistical significance at the 1%, 5%, and 10% levels, respectively.

Table 8: Return reversals by individual-investor activity

7 Short summary

- The paper studies whether investors overreact to stale financial news.
- Staleness is measured as textual similarity between a current Dow Jones story and the previous ten stories about the same firm.
- Stale news generates smaller same-day return and volume responses than fresh news, consistent with lower incremental information content.
- Nevertheless, the same-day return response to stale news predicts a reversal over the next several trading days.
- The reversal is robust to controls for known short-run return predictors, including prior returns, abnormal turnover, illiquidity, volatility, earnings-related news, and news intensity.

- Calendar-time portfolio tests produce negative post-news alphas for stale-news reactions.
- Individual investors trade more aggressively in the direction of stale news, while institutions do not show the same response.
- The reversal is stronger in stocks with above-median individual-investor trading activity.
- The main interpretation is that some investors confuse old information with new information, producing temporary price pressure.

8 Conclusion

8.1 Core conclusion

Tetlock shows that investors do not fully discount repeated information in financial news. Although stale news produces smaller immediate return and volume responses than fresh news, the remaining response is still too strong: news-day returns after stale stories partially reverse in the following week.

8.2 Economic intuition

The intuition is simple. If a story largely repeats information that already appeared, rational investors should put little weight on the repeated component. But some investors appear to react to the story as if more of it were new. This creates price pressure in the direction of the stale news. Once the market corrects this excessive reaction, returns reverse.

8.3 Incremental contribution revisited

The paper advances the literature in three main ways. First, it introduces a scalable text-based measure of news staleness. Second, it links stale-news reactions to predictable return reversals in a large sample of firm-specific news events. Third, it uses investor-type order-flow data to show that individual investors, not institutions, are the group most associated with same-direction stale-news trading.

8.4 Implications for market efficiency

The results imply that more news is not always better for short-run price efficiency. Repetition can transmit old information to investors who treat it as new. This means that financial media can both disseminate information and, at times, amplify temporary mispricing.

8.5 Implications for empirical finance

The paper shows that text can be used to measure the information environment at the event level. Similar textual tools can be used to study repeated earnings news, analyst reports, press releases, regulatory filings, social-media diffusion, and the persistence of news topics.

8.6 Limitations

The staleness measure cannot perfectly distinguish economically redundant stories from textually similar but genuinely informative updates. The order-flow data cover only one market center and a later subperiod of the sample. The empirical design is observational, so the evidence is most compelling as a combination of return predictability, robustness checks, portfolio evidence, and mechanism evidence rather than as a randomized causal experiment.

8.7 Takeaway

The enduring takeaway is that investors react not only to information content but also to information repetition. Stale news still moves prices, and those price movements reverse. Individual investors appear especially prone to treating old information as new, making stale news a source of short-run mispricing.